

Machine Learning with Python

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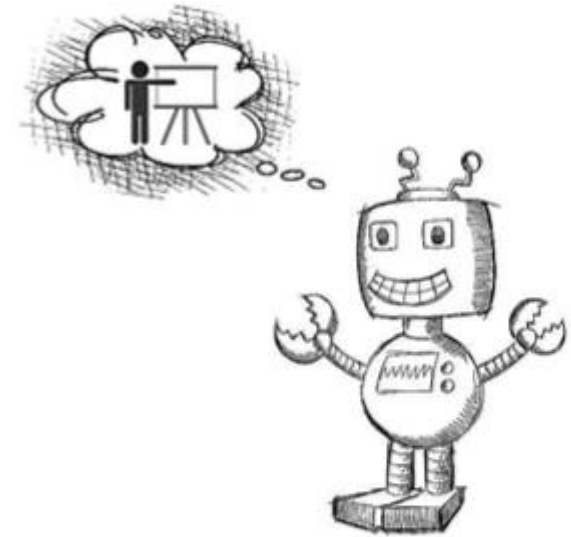
Introduction to Machine Learning and Python Setup

What is Machine Learning?

Machine learning is the field of study that focuses on computer systems that can learn from data



**HUMANS LEARN FROM
PAST EXPERIENCES**



**MACHINES FOLLOW INSTRUCTIONS
GIVEN BY HUMANS**

Machine learning trains the machines to learn specific tasks without being explicitly programmed



Cat



Dog



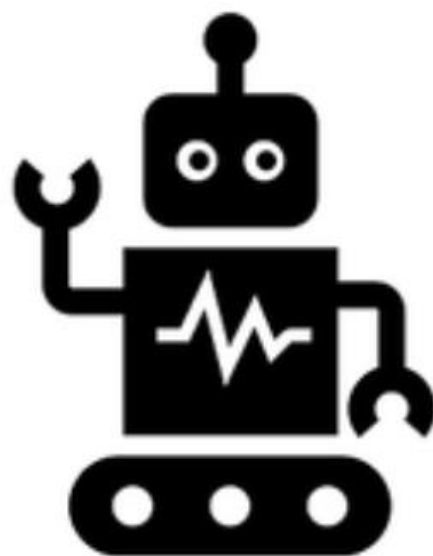
Labelled
pictures



We are **helping machine** to learn by **labeling each picture** with the name of the animal.



New picture

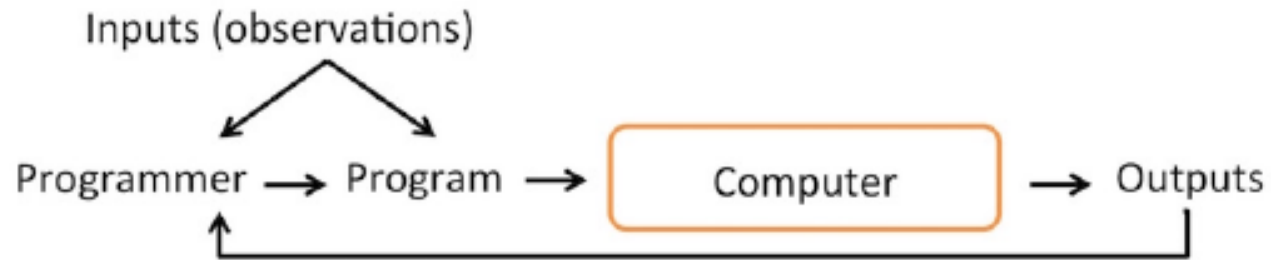


It's a Dog

The machine identified the dog even though this picture is not exactly same as any of the dog's picture shown earlier.

Machine Learning vs. Classic Programming

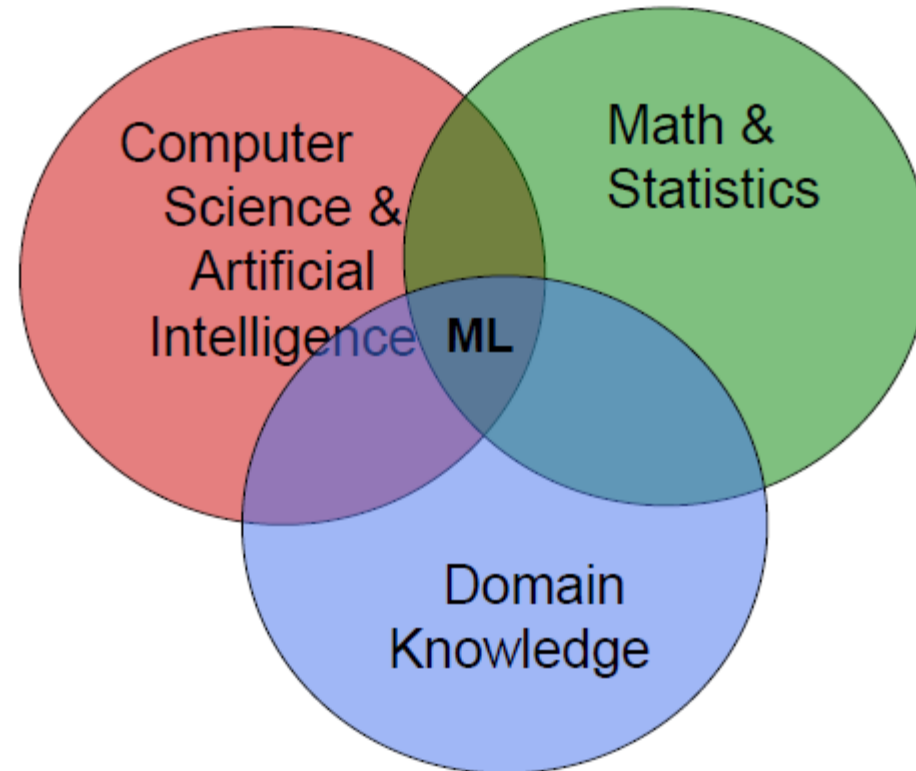
The Traditional Programming Paradigm



Machine Learning



Machine Learning (ML) is an Interdisciplinary Field



Example Application of Machine Learning

- Credit card fraud detection



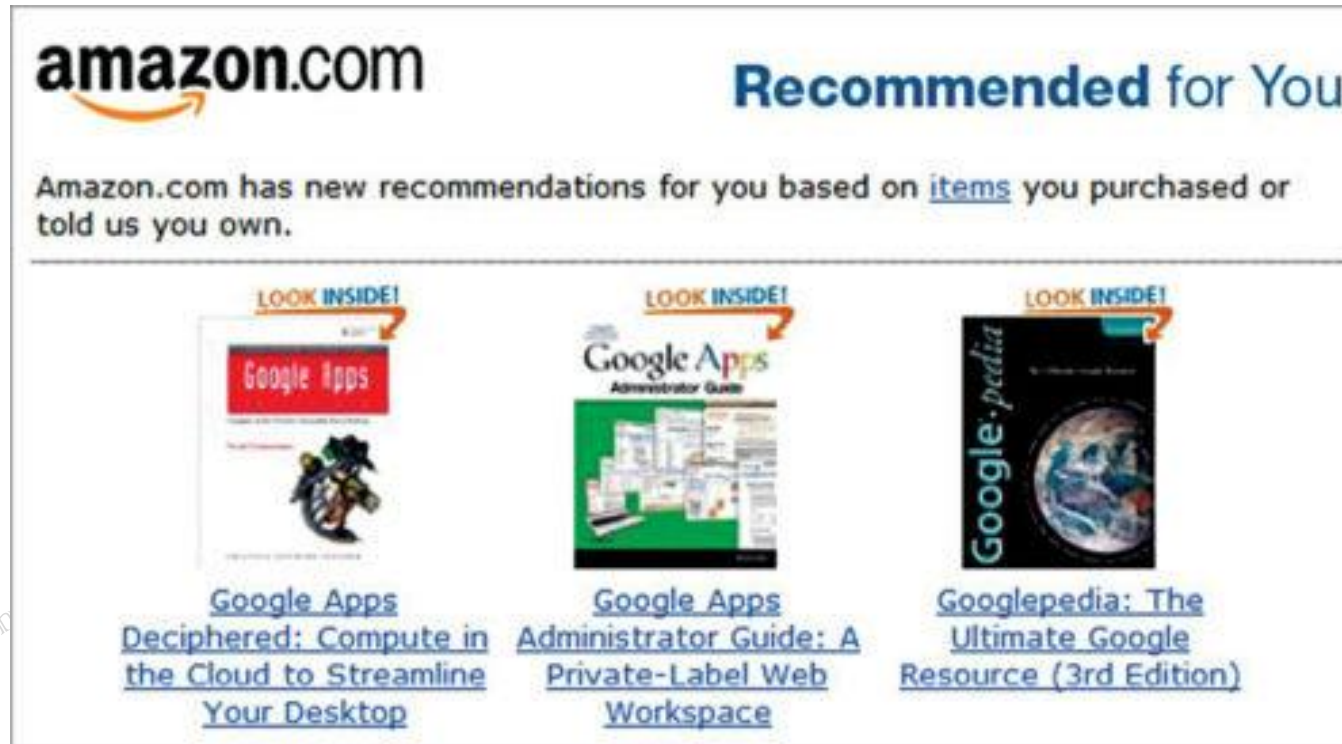
Example Application of Machine Learning

- Handwritten digit recognition

4 → 4 2 → 2 3 → 3
4 → 4 9 → 9 0 → 0
5 → 5 7 → 7 1 → 1
9 → 9 0 → 0 3 → 3
6 → 6 7 → 7 4 → 4

Example Application of Machine Learning

- Recommendations on websites



Other Applications of ML

- Targeted ads on mobile apps
- Sentiment analysis
- Climate monitoring
- Crime pattern detection
- Drug effectiveness analysis

Discussing Different Terms

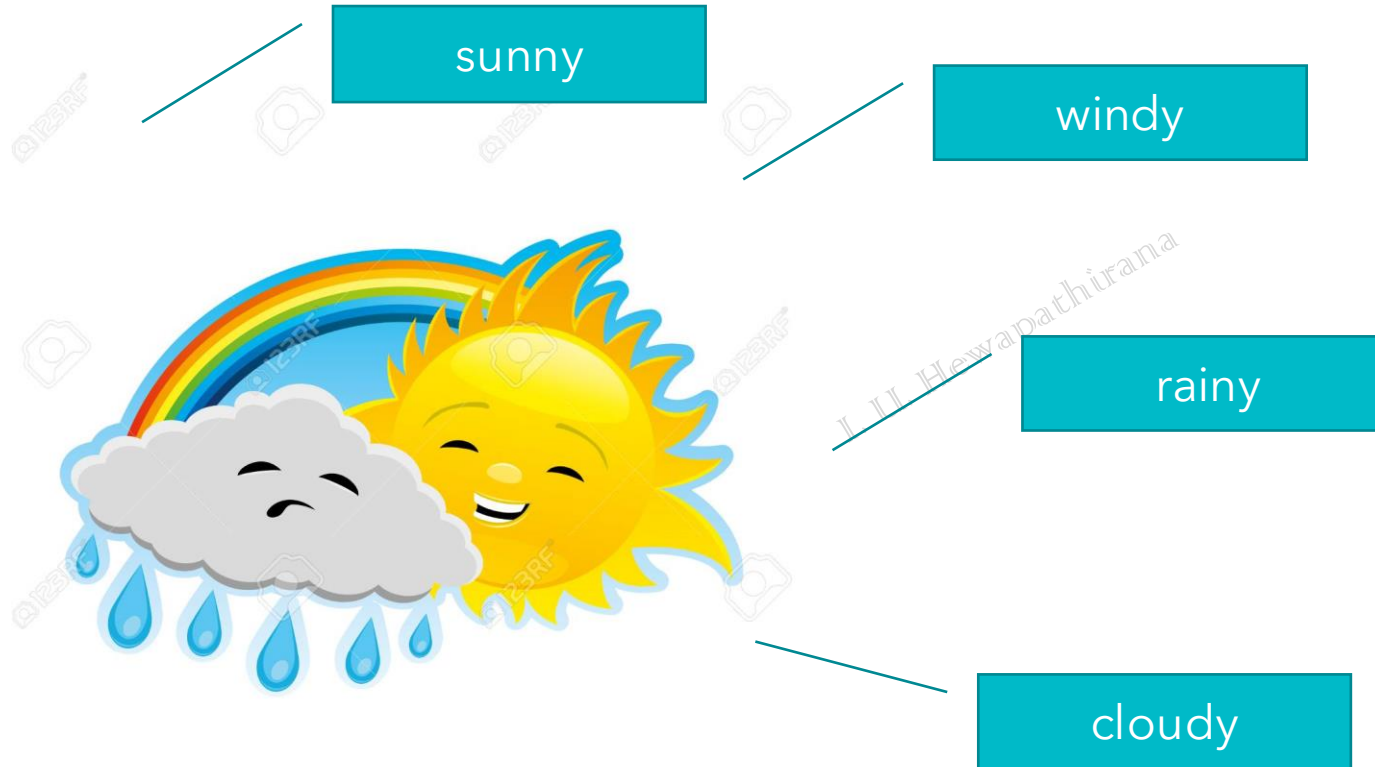
- **Machine Learning** - encompasses the algorithms and techniques used to learn from data
- **Data Mining** - activities related to finding patterns in databases and data warehouses
- **Predictive Analytics** - analyzing data in order to predict future outcomes
- **Data Science** - includes approaches for collecting, storing and managing big data

Categories of Machine Learning Techniques

- Classification
- Regression
- Cluster Analysis
- Association Analysis

Classification

Goal: Predict Category



Classification Examples

- Classify tumor as benign or malignant
- Predict if it will rain tomorrow
- Determine if loan application is high-, medium-, or low-risk
- Identify sentiment as positive, negative, or neutral

Regression



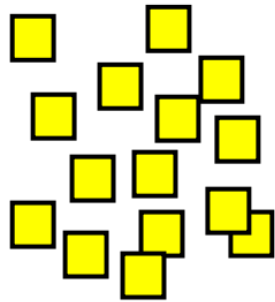
Goal: Predict Numeric Value

Regression Examples

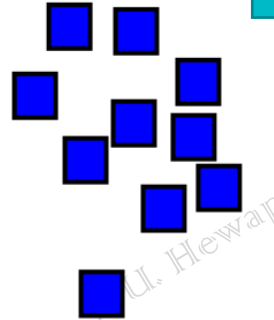
- Estimate demand for a product based on time of year
- Predict score on a test
- Determine likelihood of drug effectiveness for patient
- Predict amount of rain

Cluster Analysis

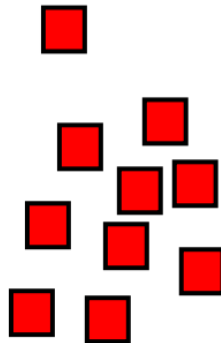
High-Value Shoppers



Occasional Buyers



Bargain Hunters



Goal: Organize similar items into groups

Cluster Analysis Examples

- Identify areas of similar topography (desert, grass, etc.)
- Categorize different types of tissues from medical images
- Determine different groups of weather patterns
- Discover crime hot spots

Association Analysis

Goal: Find rules to capture associations between items



Association Analysis Examples

- Recommend items based on purchase/browsing history
- Have sales on related items often purchased together
- Identify web pages accessed together

Categories of Machine Learning

Supervised Learning

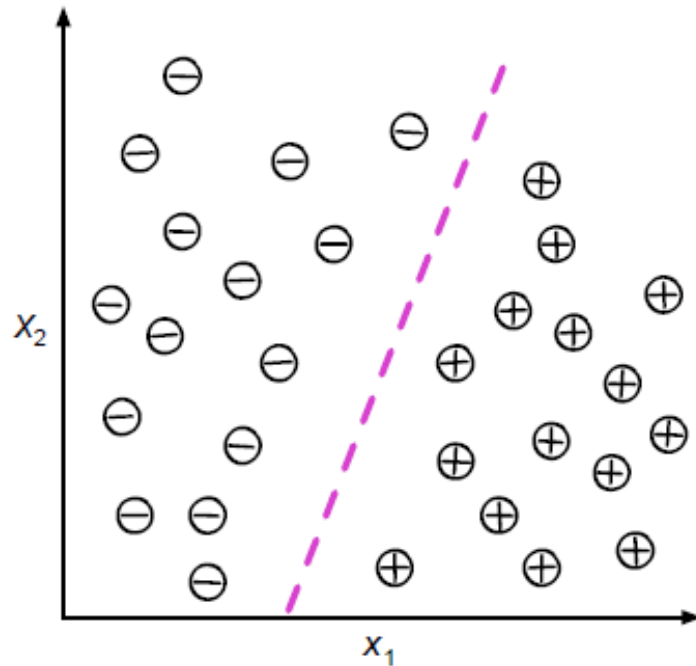
- Labeled data
- Direct feedback
- Predict outcome/future

Unsupervised Learning

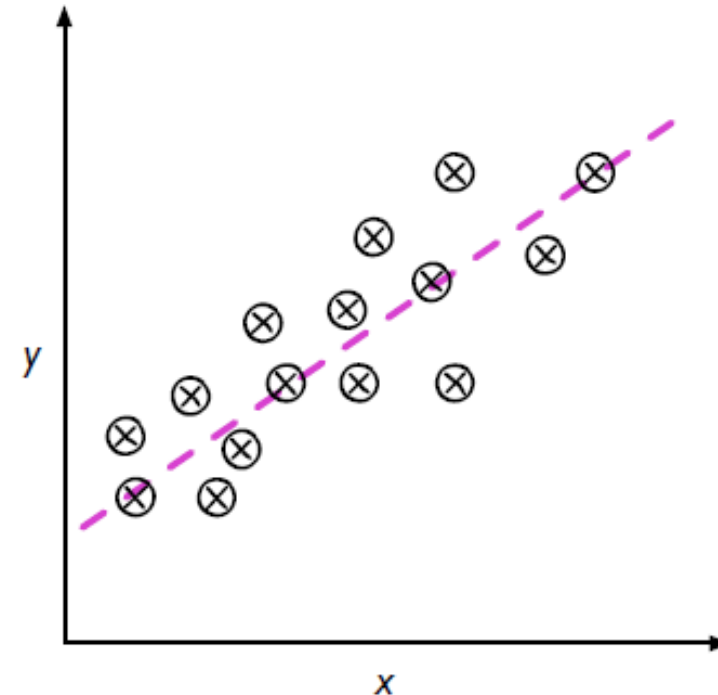
- No labels/targets
- No feedback
- Find hidden structure in data

Supervised Learning

Classification

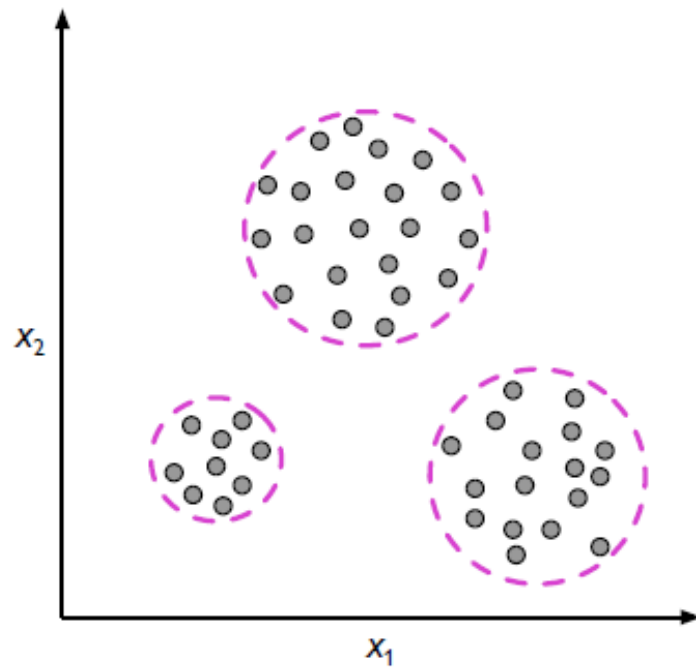


Regression



Unsupervised Learning

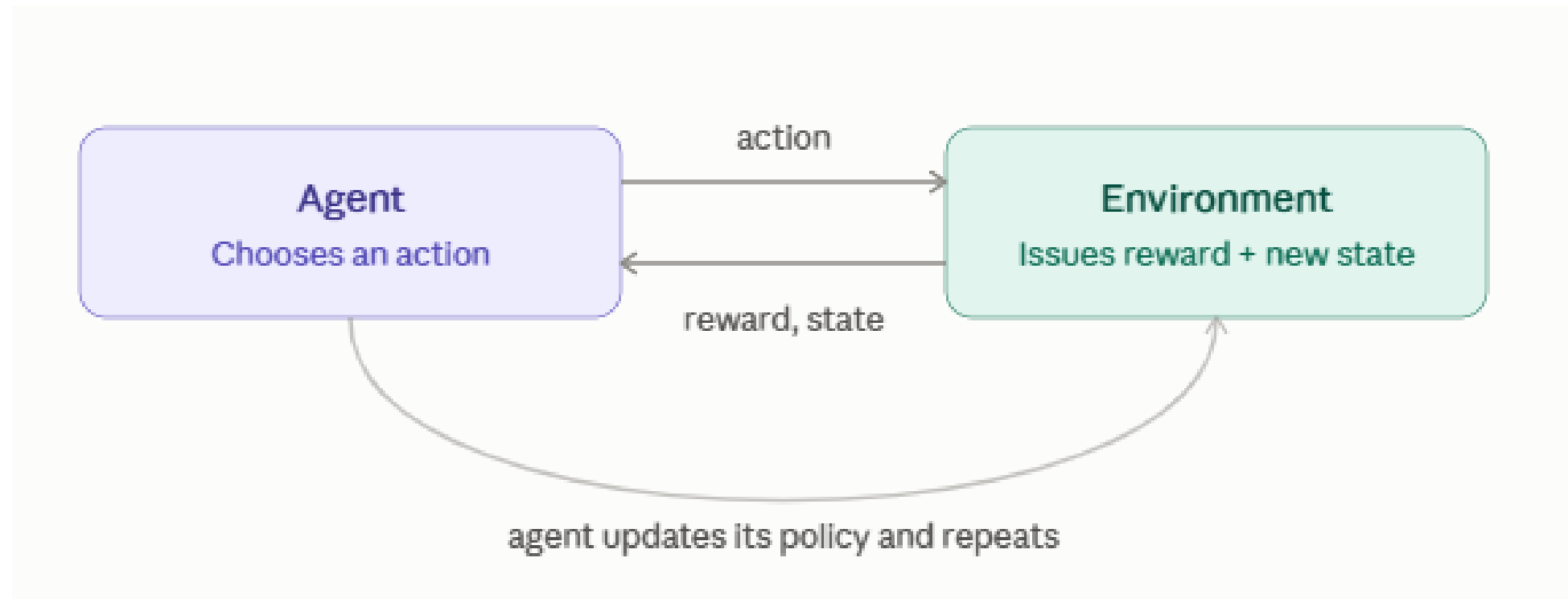
Clustering



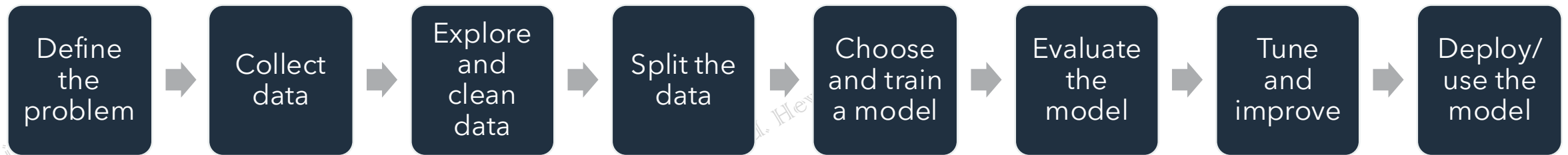
Association Analysis



Reinforcement Learning



The Machine Learning Process



Key Terminology

Term	Meaning
Feature	An input variable used to make predictions (e.g., square footage of a house)
Label / Target	The output value the model is trying to predict (supervised learning only)
Model	The mathematical representation learned from data, used to make predictions
Training	The process of fitting a model to data
Training / Test set	Data used to train the model / data held back to evaluate it
Overfitting	When a model learns the training data too closely and fails to generalize to new data
Hyperparameter	A setting chosen before training that controls how the model learns

Course Roadmap Preview

- **Day 1:** ML, Python and Google Colab foundations
- **Day 2:** Data wrangling and Regression (Supervised Learning – predicting numbers)
- **Day 3:** Classification (Supervised Learning – predicting categories) using different algorithms
- **Day 4:** Hyperparameter tuning, Unsupervised Learning